

T Individual cattle as agents and their behavior in healthy and disease states in an integrated beef production system: An agent-based simulation approach

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Bovine Respiratory Disease (BRD) is the leading cause of cattle morbidity and mortality in beef production systems in the United States, with an economic burden of up to \$2 billion annually. BRD emerges from complex interactions among multiple factors, including environmental stressors, cow-to-cow interactions, management practices, and more. The use of antimicrobials is common, but overuse risks selection for pathogen resistance, while underuse may allow outbreaks to spread if unchecked. This constitutes a major challenge to various stakeholders within the beef production value chain, as cattle infected with the disease lose performance, utilize treatment resources, and take longer to reach their market weight, hence decreasing the overall productivity and profitability of the industry. To address this issue and enhance productivity, our team designed an agent-based model that simulates disease transmission across cow-calf, stocker, and feedlot phases using an SEIR framework with quarantine controls. The model integrates epidemiological and economic outcomes to enable stakeholders to evaluate intervention strategies and forecast how vaccination and movement restrictions affect productivity and profitability in the beef production system.

This study used NetLogo v6.4 to create a simulation in which cattle agents progress through the beef production chain from cow-calf to feedlot stages with heterogeneous characteristics, random behaviors, defined attributes, and decision rules (Figure 1). The model integrates a set of mathematical equations to estimate key factors influencing the spread of BRD as well as changes in cattle weight and economic cost. Each agent interacts with its environment according to defined decision rules, allowing the simulation to capture how BRD spreads within an interconnected production system. Scenario analysis was conducted to observe the outcomes of various management decisions on production dynamics.

Our quantitative agent-based model demonstrated that targeted interventions such as vaccination, informed metaphylaxis, and effective quarantine can substantially mitigate the burden of BRD and optimize profit through reduced production losses related to changes in average daily gain (ADG) and time to recovery. In conclusion, the sample paths generated by our ABM can be used to promote optimal decision-making during a BRD epidemic, and the generated data/insights can be used to predict accrued losses from disease events.

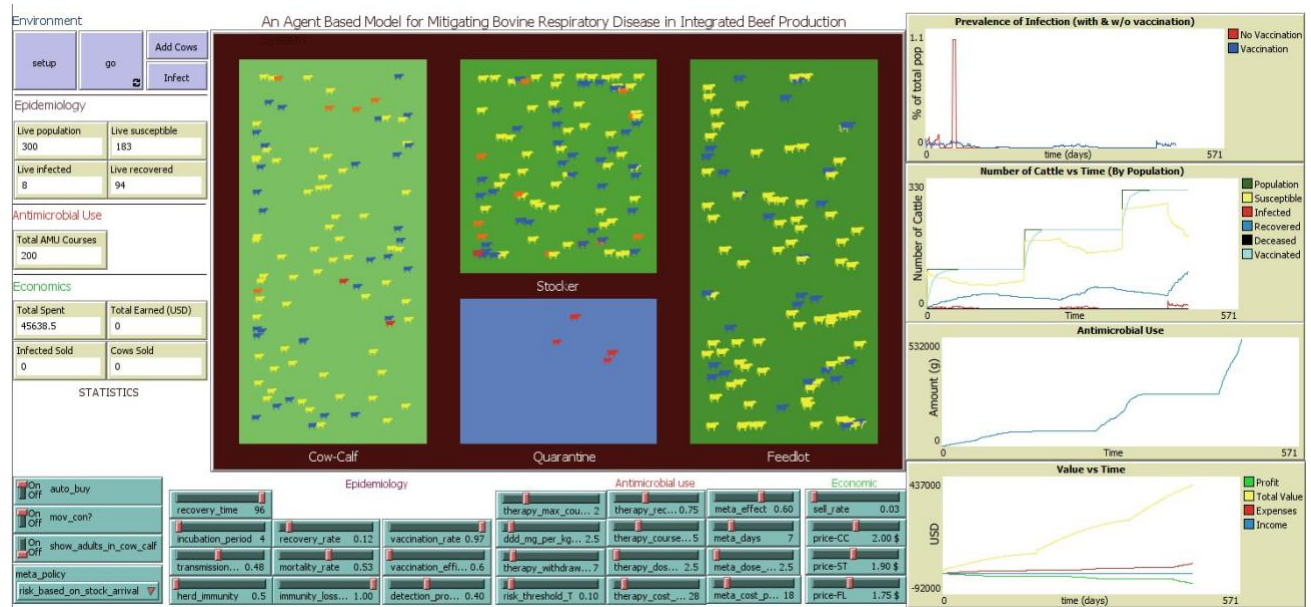


Figure 1. A Netlogo interface showing an agent-based model for the mitigation of BRD in an integrated beef production system.